The Inflation Risk Premium in Zimbabwean Government Bonds

Lord Soumadeep Ghosh with ChatGPT (OpenAI)

Kolkata, India

Abstract

This paper develops an interdisciplinary framework for investigating the inflation risk premium in Zimbabwean government bonds. It combines asset pricing, monetary economics, sovereign debt analysis, econometrics, statistical inference, computational methods, institutional economics, and welfare analysis. Zimbabwe's history of inflation, monetary regime changes, exchange-rate instability, dollarization, and fiscal financing pressures makes the identification of inflation compensation particularly challenging. Recent IMF assessments document both periods of monetary instability and subsequent stabilization following monetary tightening [9, 10].

The central proposition is that observed nominal government-bond yields cannot be interpreted as pure inflation compensation. Expected inflation, inflation risk, real interest rates, sovereign credit risk, liquidity, currency denomination, and institutional credibility must be distinguished. The paper proposes a latent-state econometric framework, explicit hypotheses, a reproducible estimation algorithm, and a policy-oriented interpretation. It does not report an estimated Zimbabwean inflation risk premium because a matched, denomination-consistent dataset of bond prices, inflation expectations, and risk controls has not been assembled here.

The paper ends with "The End"

Keywords: Inflation risk premium; Zimbabwe; government bonds; sovereign risk; ZiG; dollarization; monetary credibility; econometrics; asset pricing.

1 Introduction

The inflation risk premium (IRP) compensates investors for exposure to uncertainty about future inflation. It is conceptually distinct from expected inflation, although both may contribute to the yield on a nominal government bond. The distinction follows from modern asset-pricing theory and the analysis of nominal interest rates [8, 4, 2].

Zimbabwe provides a particularly important setting for this investigation. The economy has experienced repeated episodes of high inflation, currency instability, changes in monetary regimes, and restricted access to international financing. The introduction of Zimbabwe Gold (ZiG) in April 2024 was followed by substantial monetary expansion and currency depreciation in September 2024. Subsequent monetary tightening helped stabilize the exchange rate and reduce monthly inflation during the first half of 2025 [9, 10].

These developments create a natural setting for studying how inflation uncertainty, monetary credibility, sovereign financing needs, and market segmentation interact. However, historical instability alone does not establish the sign or magnitude of the IRP. That requires empirical identification.

The paper addresses five questions:

1. How should the IRP be defined for Zimbabwean government securities?

2. How can inflation compensation be separated from sovereign, liquidity, currency, and term premia?

3. Which econometric framework can accommodate structural breaks and changes in monetary regimes?

4. How should the resulting estimates be interpreted for monetary policy, debt management, and social welfare?

5. How can the analysis be made reproducible and falsifiable?

The guiding methodological principle is that shocks may disturb economic systems, but constructive institutional reform and diplomacy are preferable outcomes to persistent instability.

2 Theoretical framework

2.1 Nominal yields and inflation compensation

Let $y_{t,n}^{N,c}$ denote the nominal yield at time t on a government bond with maturity n, denominated in currency c. A useful approximate decomposition is

$$y_{t,n}^{N,c} = \overline{r}_{t,n}^{e,c} + \overline{\pi}_{t,n}^{e,c} + IRP_{t,n}^{c} + TP_{t,n}^{c} + CRP_{t,n}^{c} + LP_{t,n}^{c} + RP_{t,n}^{c}. \tag{1}$$

Here, $\overline{r}^{\,e}$ is the expected average real short rate, $\overline{\pi}^{\,e}$ is expected average inflation in the relevant currency, IRP is the inflation risk premium, TP is the residual term premium, CRP is compensation for sovereign credit risk, LP is a liquidity premium, and RP collects other relevant risk and market-segmentation effects.

Equation (1) is an approximate empirical accounting relation, not a unique structural identity. Its components depend on the model, information set, and security characteristics. In particular, the decomposition must avoid double-counting risks already incorporated into the estimated term or sovereign premia [6, 1].

The Fisher relation provides a useful starting point. For a one-period nominal return, the exact gross-return relationship is

$$1 + i_t = (1 + r_t)(1 + \pi_{t+1}), \tag{2}$$

where i_t is the nominal return, r_t is the real return, and π_{t+1} is realized inflation.

Taking logarithms and expectations gives an approximate relation between nominal yields, expected real returns, and expected inflation. The difference between nominal yields and expected inflation is not necessarily a pure real interest rate because risk compensation and market frictions remain.

2.2 A pricing interpretation

Let M_{t+1} denote the stochastic discount factor and R_{t+1}^{N} the gross nominal return on an asset. Under standard asset-pricing conditions,

$$1 = \mathbb{E}_t \left[M_{t+1} R_{t+1}^{N} \right]. \tag{3}$$

Consequently,

$$\mathbb{E}_t[R_{t+1}^{N}] = \frac{1}{\mathbb{E}_t[M_{t+1}]} - \frac{\mathrm{Cov}_t(M_{t+1}, R_{t+1}^{N})}{\mathbb{E}_t[M_{t+1}]}. \tag{4}$$

This expression illustrates how expected returns depend on covariance with the stochastic discount factor. Inflation risk commands compensation when the relevant exposure is costly

to investors in states where marginal utility is high. The sign of the resulting premium is not determined by inflation volatility alone [5, 4].

2.3 Currency denomination and sovereign risk

Zimbabwe's monetary history makes currency denomination central to identification. A ZiG-denominated bond and a United States dollar-denominated bond issued by the same sovereign are different financial instruments.

Their yields reflect different inflation processes, currency risks, investor bases, liquidity conditions, and potentially different repayment arrangements. Therefore, the empirical analysis must not subtract United States inflation expectations from a ZiG nominal yield or treat exchange-rate depreciation as identical to domestic inflation.

For a foreign-currency bond, the relevant decomposition should use inflation expectations associated with that currency and account separately for sovereign credit, liquidity, and any currency-conversion exposure borne by the investor.

3 Zimbabwe's institutional and macroeconomic setting

3.1 Monetary instability and stabilization

The IMF's 2025 Article IV assessment describes how rapid expansion of the ZiG monetary base preceded a sharp depreciation in September 2024 and a subsequent inflation spike. Tighter monetary policy, higher reserve requirements, and the cessation of monetary financing helped stabilize the exchange rate and reduce monthly inflation by June 2025 [9, 10].

This sequence motivates a regime-dependent rather than constant-premium model. Nevertheless, it does not independently identify the IRP: monetary tightening can simultaneously affect expected inflation, real rates, liquidity, default concerns, and investor risk appetite.

3.2 Public debt and financial intermediation

The IMF's debt sustainability analysis reports a substantial increase in domestic public debt and a concentration of Treasury bonds among domestic banks and a major private asset manager. It also identifies risks arising from the bank–sovereign nexus and the crowding out of private lending [10].

These features matter for bond pricing. If financial institutions hold government securities for regulatory, liquidity-management, or institutional reasons, observed yields may not be generated by a fully competitive market with diverse investors. Yield movements may consequently reflect market segmentation and financial repression as well as inflation risk.

Table 1: Institutional features and their empirical implications

Feature	**Pricing channel**	**Empirical implication**
Monetary instability	Inflation expectations and risk	Allow time-varying inflation states
Currency depreciation	Currency and imported-price exposure	Model exchange rates separately
Fiscal financing pressure	Sovereign and monetary credibility risk	Include fiscal and debt indicators
Concentrated bond ownership	Segmentation and limited price discovery	Measure market concentration where possible
Dollarization	Multiple currency-specific pricing processes	Estimate separate denomination-specific models
Structural monetary reforms	Changes in policy credibility	Test for breaks and regime changes

4 Identification and testable hypotheses

The principal identification problem is that the IRP is latent. A nominal yield alone cannot separately identify expected inflation, inflation risk, real rates, and other premia.

Break-even inflation is also not a direct measure of expected inflation unless differences in liquidity, credit risk, taxation, and other relevant characteristics are negligible or adequately controlled. This is a general limitation of extracting inflation expectations from nominal and inflation-linked securities [4, 2].

Zimbabwean data may not provide a sufficiently liquid, maturity-matched market in inflation-linked securities. In that case, the IRP should be estimated as a model-dependent latent component, with uncertainty reported explicitly.

The following hypotheses are proposed.

H_1**:** The IRP varies over time and across monetary regimes.

H_2**:** Greater inflation uncertainty is associated with changes in the estimated IRP, conditional on real rates, sovereign risk, and liquidity.

H_3**:** Measures of fiscal financing pressure and monetary-policy credibility improve the explanatory power of the yield decomposition.

H_4**:** Estimates obtained without liquidity and sovereign-risk controls differ materially from estimates obtained with those controls.

H_5**:** The IRP differs across currency denominations and maturity horizons.

These are empirical hypotheses, not established findings about Zimbabwe. Their validity must be assessed using appropriate data, robustness tests, and uncertainty intervals.

5 Econometric methodology

5.1 Latent-state representation

A parsimonious state-space model can represent the unobserved components of nominal yields. Let z_t contain the latent inflation-risk state and other latent pricing factors. The measurement equation is

$$y_t = X_t\beta + Hz_t + \varepsilon_t, \tag{5}$$

where X_t contains observed controls, β is a vector of coefficients, H maps latent states into observed yields, and ε_t is measurement error.

The state-transition equation is

$$z_t = A_{s_t} z_{t-1} + B_{s_t} u_t + \eta_t, \tag{6}$$

where s_t denotes the monetary regime, u_t contains observed macroeconomic innovations, and η_t represents state innovations.

The regime indicator may distinguish periods of monetary expansion, exchange-rate adjustment, stabilization, and subsequent policy changes. Regime dates should be based on documented institutional events or estimated using a clearly specified statistical procedure.

A structural-break model is appropriate when parameter stability cannot reasonably be assumed [3]. State-space methods and the Kalman filter provide a computational framework for estimating latent variables under suitable linear-Gaussian assumptions [7].

5.2 Identification restrictions

A latent-state model does not automatically identify the IRP. Multiple decompositions may fit the same yield data.

Identification requires additional information or restrictions, such as:

1. Survey-based inflation expectations at relevant horizons.
2. Matched nominal and inflation-linked securities, where available and sufficiently liquid.
3. External estimates of expected real rates.
4. Sovereign credit-risk indicators and debt-service data.
5. Bid–ask spreads, turnover, issuance, and other liquidity measures.
6. Explicit restrictions on state dynamics and measurement errors.

If inflation-linked securities or reliable survey expectations are unavailable, the analysis must report the resulting identification limitations rather than manufacture precision.

5.3 Statistical inference

The baseline null hypothesis is

$$H_0 : IRP_t = 0 \quad \text{for all } t \text{ in the estimation sample.} \tag{7}$$

A useful alternative allows the premium to vary across time or regimes. A zero estimate should not automatically be interpreted as proof that inflation risk is economically irrelevant. It may also reflect offsetting risks, limited information, or weak identification.

Inference should incorporate parameter uncertainty, measurement error, model sensitivity, and potential structural breaks. Where asymptotic approximations are unreliable, bootstrap procedures or simulation-based inference may be appropriate. Out-of-sample evaluation should compare predictive performance against transparent benchmark models.

6 Data and reproducibility

The preferred dataset is a security-level panel containing government-bond prices or yields, maturities, issue dates, redemption terms, currency denominations, and transaction or valuation dates.

It should be joined to inflation, exchange-rate, fiscal, monetary, and financial-market data. The IMF's country report provides important institutional context but does not substitute for security-level observations or an independently identified inflation risk premium [9, 10].

Table 2: Proposed data architecture

Data block	Variables	Purpose
Government bonds	Prices, yields, maturities, currency	Measure nominal financing costs
Prices and expectations	CPI, forecasts, survey expectations	Separate expected inflation from risk
Foreign exchange	Official and parallel-market rates, where reliable	Measure currency instability
Monetary policy	Policy rates, reserve requirements, monetary aggregates	Capture policy stance and regime changes
Fiscal accounts	Deficit, debt, arrears, issuance	Control for sovereign financing pressure
Market liquidity	Turnover, bid–ask spreads, issue concentration	Estimate liquidity and segmentation effects
External conditions	Commodity prices, external rates, remittances	Control for common and external shocks

Candidate sources include the Reserve Bank of Zimbabwe, Zimbabwe National Statistics Agency, Ministry of Finance, IMF, World Bank, and relevant securities-market records. The availability, frequency, currency consistency, and historical coverage of each series must be verified before estimation.

The empirical dataset should retain source metadata, observation dates, revisions, currency units, transformations, and missing-value indicators. No interpolation should be applied without documenting its assumptions.

7 Computational procedure

The following pseudocode specifies a reproducible estimation workflow. It deliberately separates data validation, model estimation, and interpretation.

```
INPUT:
Security-level government-bond data
Inflation and inflation-expectation data
Exchange-rate and monetary-policy data
Fiscal, sovereign-risk, and liquidity controls

1. Validate source metadata, dates, units, and currencies.
2. Separate securities by currency denomination and maturity.
3. Construct matched observation windows.
4. Inspect missingness, outliers, and measurement breaks.
5. Define baseline and alternative identification restrictions.
```

```
6. Estimate the yield decomposition and latent-state model.
7. Test for structural breaks and regime dependence.
8. Compute confidence or credible intervals for the IRP.
9. Compare estimates across model specifications.
10. Evaluate forecasts using a chronological holdout sample.
11. Save estimates, diagnostics, code versions, and metadata.
12. Report limitations and avoid unsupported causal claims.

OUTPUT:
Estimated IRP paths, uncertainty intervals, diagnostics,
robustness comparisons, and reproducible research records.
```

A valid implementation should test whether the model recovers known parameters on simulated data, detect singular or weakly identified specifications, and verify that results are reproducible under fixed random seeds and documented software versions.

Machine-learning methods may help forecast yields, volatility, or inflation expectations. However, predictive accuracy does not by itself establish a structural interpretation of the IRP. A transparent econometric benchmark should therefore precede more flexible algorithms.

8 A multidisciplinary interpretation

8.1 Economics and finance

The economic interpretation begins with the distinction between expected compensation and risk compensation. Nominal yields reflect the price of borrowing, but they also embody investors' expectations about future purchasing power and their exposure to uncertain economic states.

Sovereign financing costs can also rise when fiscal credibility weakens or investors doubt repayment arrangements. These effects must not be mechanically assigned to the IRP.

8.2 Physics-inspired modeling and statistical mechanics

A regime-switching state-space model provides a useful analogy with physical systems whose observable outcomes depend on latent states. Inflation, exchange-rate stability, and policy credibility may jointly define a changing macroeconomic state.

The analogy is methodological rather than ontological: economic agents learn, anticipate policy, and respond strategically. Economic regimes therefore cannot be assumed to obey physical laws or fixed conservation equations.

8.3 Psychology, psychiatry, and behavioral finance

Inflation expectations may be affected by memory, loss aversion, adaptive learning, uncertainty, and trust in institutions. Historical episodes of monetary instability may influence how households and investors evaluate promises of monetary stability.

These mechanisms suggest testable extensions involving surveys, expectation disagreement, and measures of institutional trust. They do not justify inferring a psychiatric condition from market behavior or treating an entire population as psychologically homogeneous.

8.4 Political science, geopolitics, and diplomacy

Monetary credibility is partly institutional. Fiscal coordination, central-bank independence, public financial management, and predictable legal arrangements influence the environment in which sovereign securities are priced.

External financing restrictions, commodity-price shocks, and international relations can also affect fiscal capacity and exchange-rate expectations. The IMF identifies external financing constraints and domestic fiscal pressures as important elements of Zimbabwe's macroeconomic outlook [9, 10].

Diplomatic engagement and credible debt-resolution arrangements may improve access to financing, although the magnitude and timing of any effect on the IRP must be established empirically.

8.5 Welfare economics and warfare economics

A sovereign yield is not merely a financial statistic. It can affect the government's cost of financing public services, infrastructure, health, education, and social protection.

High borrowing costs may crowd out productive investment. Conversely, artificially suppressed yields may transfer risks to banks, pension funds, or households and distort capital allocation. Welfare analysis must therefore consider distributional effects rather than treating lower nominal yields as an unqualified social improvement.

Defense spending and other security expenditures may compete with development expenditure under a constrained fiscal envelope. The appropriate analysis compares their marginal social benefits, opportunity costs, financing risks, and long-run consequences. It does not presume that higher military expenditure is either necessary or welfare-enhancing.

8.6 Chemistry, biology, and public health

These fields enter the analysis through real economic transmission channels rather than through a direct chemical or biological determination of bond premia. Energy prices, agricultural shocks, food supply, disease burdens, and public-health emergencies can affect productivity, fiscal expenditure, inflation, and household welfare.

Their inclusion is justified when observable indicators improve the economic model. They should not be added as decorative variables without a causal mechanism or an empirical measurement strategy.

9 Conceptual diagram

Figure 1 summarizes the proposed relationship between macroeconomic conditions, bond pricing, latent risk components, and policy outcomes.

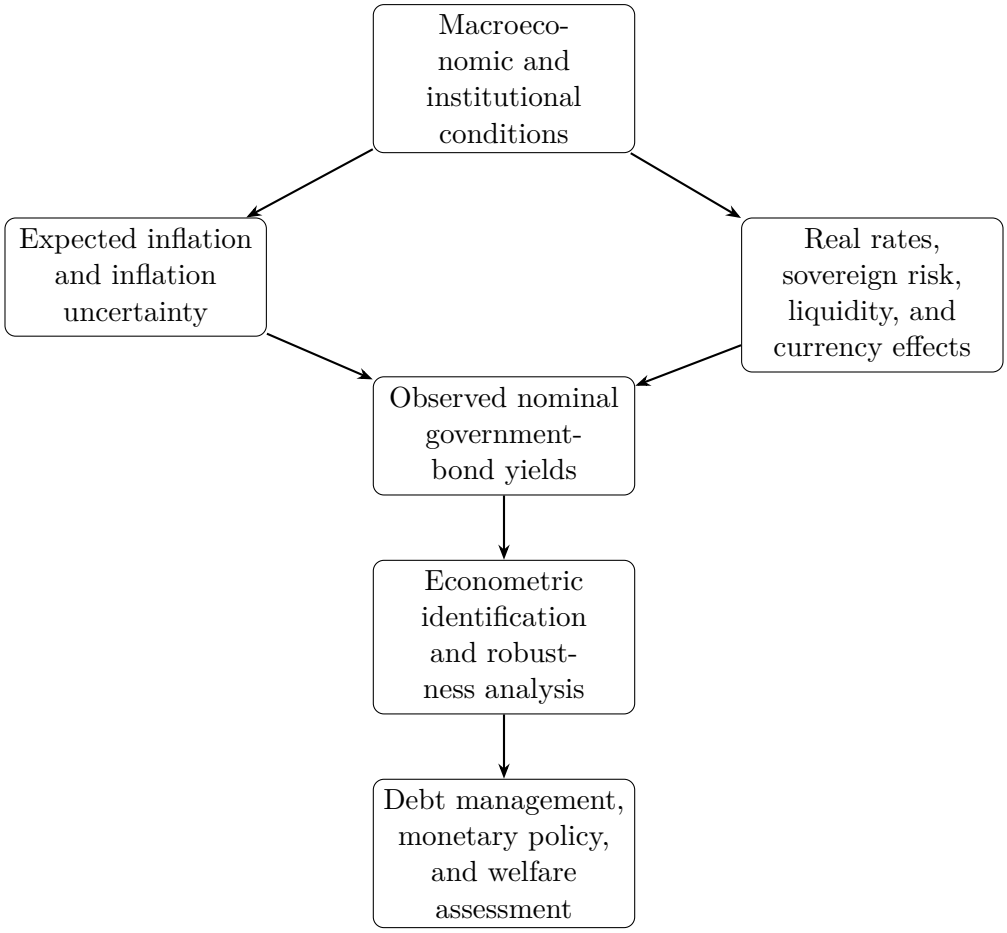


Figure 1: Conceptual framework for identifying the inflation risk premium in Zimbabwean government bonds.

10 Policy implications

The framework yields several conditional policy implications.

1. **Improve fiscal transparency.** Reliable debt, arrears, and issuance data help investors distinguish sovereign financing risk from inflation risk.
2. **Strengthen monetary credibility.** Predictable monetary arrangements can improve the formation of inflation expectations, although their effects on the IRP must be measured rather than assumed.
3. **Improve market transparency.** Consistent security-level prices, transaction data, and publication of issuance terms would improve price discovery.
4. **Account for currency segmentation.** ZiG-denominated and foreign-currency securities should be analyzed separately before making cross-currency comparisons.
5. **Protect financial intermediation.** Concentrated sovereign exposures and high-cost short-term financing can create vulnerabilities for banks and restrict productive private credit [10].
6. **Evaluate social consequences.** Policy evaluation should consider household purchasing power, public-service provision, investment, and distributional outcomes alongside government borrowing costs.

These recommendations are consistent with the broader objective of restoring durable macroeconomic stability. They do not imply that any single reform will necessarily reduce the estimated IRP by a predetermined amount.

11 Limitations and research agenda

Four limitations deserve particular emphasis.

First, the IRP is not directly observed. Its estimated value depends on the availability of inflation expectations, the identification of other risk premia, and the maintained pricing model.

Second, Zimbabwe's multiple monetary and currency arrangements make consistent historical comparisons difficult. Currency redenominations, changes in price measurement, and structural breaks require explicit treatment.

Third, thin trading, concentrated bond ownership, and possible administrative influences on yields can weaken the interpretation of observed prices as market-clearing prices.

Fourth, the macroeconomic developments documented by the IMF through its 2025 assessments cannot substitute for an updated security-level dataset. The numerical results required to accept or reject the proposed hypotheses remain to be estimated.

Future research should therefore prioritize data construction, identification diagnostics, regime-sensitive estimation, and independent replication. Results should be reported across alternative specifications rather than selected solely because they support a preferred theoretical conclusion.

12 Conclusion

The inflation risk premium in Zimbabwean government bonds is a latent financial quantity embedded within a broader system of monetary, fiscal, institutional, and market risks.

The central methodological contribution of this paper is to separate the definition of the IRP from the empirical problem of identifying it. A high nominal yield is not sufficient evidence of a high inflation risk premium, and a period of declining inflation does not establish that the premium has disappeared.

A credible empirical investigation requires currency-consistent bond data, defensible inflation expectations, sovereign-risk controls, liquidity measures, explicit structural-break treatment, and transparent uncertainty estimates.

The broader objective is not merely to explain sovereign yields. It is to develop evidence that can support sound public finance, financial stability, productive investment, and improvements in social welfare. Mathematical rigor and empirical honesty should remain complementary to institutional reform and constructive international cooperation.

References

[1] Adrian, T., Crump, R. K., and Moench, E. (2013). Pricing the term structure with linear regressions. *Journal of Financial Economics*, 110(1), 110–138.

[2] Ang, A., Bekaert, G., and Wei, M. (2008). The term structure of real rates and expected inflation. *The Journal of Finance*, 63(2), 797–849.

[3] Bai, J., and Perron, P. (2003). Computation and analysis of multiple structural change models. *Journal of Applied Econometrics*, 18(1), 1–22.

[4] Campbell, J. Y., and Shiller, R. J. (1996). A scorecard for indexed government debt. In B. S. Bernanke and J. J. Rotemberg (eds.), *NBER Macroeconomics Annual 1996*, Volume 11, pp. 155–208. MIT Press.

[5] Cochrane, J. H. (2005). *Asset Pricing*, revised edition. Princeton University Press.

[6] Duffie, D., and Singleton, K. J. (1999). Modeling term structures of defaultable bonds. *The Review of Financial Studies*, 12(4), 687–720.

[7] Durbin, J., and Koopman, S. J. (2012). *Time Series Analysis by State Space Methods*, second edition. Oxford University Press.

[8] Fisher, I. (1930). *The Theory of Interest*. Macmillan.

[9] International Monetary Fund (2025). Zimbabwe: 2025 Article IV Consultation— Press Release; Staff Report; and Statement by the Executive Director for Zimbabwe. IMF Country Report No. 25/282, published October 2, 2025. Available at: https://www.imf.org/en/publications/cr/issues/2025/10/02/zimbabwe-2025-article-iv-consultation-press-release-staff-report-and-statement-by-the-5

[10] International Monetary Fund (2025). Zimbabwe: Staff Report for the 2025 Article IV Consultation— Debt Sustainability Analysis. In IMF Country Report No. 25/282, published October 2, 2025. Available at: https://www.elibrary.imf.org/view/journals/002/2025/282/article-A002-en.xml.

Glossary

Inflation risk premium (IRP)
Compensation associated with exposure to uncertainty about future inflation.

Expected inflation
The conditional expectation of future inflation over a specified horizon.

Nominal yield
The yield on a security expressed in nominal currency units.

Real interest rate
The interest rate measured in purchasing-power terms.

Sovereign credit risk
The risk that a government will fail to meet its contractual debt obligations.

Liquidity premium
Compensation associated with the difficulty or cost of trading a security without materially affecting its price.

Term premium
Compensation associated with holding a longer-maturity security rather than rolling over shorter-maturity securities, under a specified term-structure model.

Dollarization
The use of a foreign currency alongside, or instead of, domestic currency for transactions, saving, or financial contracts.

Structural break
A change in the statistical properties or parameters of a process.

Latent state
An unobserved variable inferred from observable data through a statistical model.

State-space model
: A statistical framework comprising a measurement equation and a latent-state transition equation.

Financial repression
: Institutional or regulatory arrangements that constrain financial prices, portfolio choices, or capital allocation.

Identification
: The conditions under which model parameters or latent economic quantities can be uniquely inferred from data.

Welfare economics
: The study of how resource allocation and economic policies affect social welfare and its distribution.

The End